

AE 6114 - Fundamentals of Solid Mechanics
Fall 2022

Homework 4: 2D Linear Elasticity

Problem 1

1. Consider the following 2D stress distribution on a body:

$$\sigma_{11} = 12x_1^4x_2^2, \quad \sigma_{22} = 12x_1^2x_2^4, \quad \sigma_{12} = -16x_1^3x_2^3$$

- (a) Show that this stress distribution satisfies the equation of equilibrium when no body forces are present.
- (b) Assuming isotropic linear elastic material, find the corresponding 2D strains ϵ_{11} , ϵ_{12} , and ϵ_{22} .
- (c) Show that these strains do not satisfy the compatibility equation

$$\frac{\partial^2 \epsilon_{11}}{\partial x_2^2} + \frac{\partial^2 \epsilon_{22}}{\partial x_1^2} = 2 \frac{\partial^2 \epsilon_{12}}{\partial x_1 \partial x_2}$$

- (d) Integrate the ϵ_{11} strains to obtain an expression for the displacement component u_1 and the ϵ_{22} strains to obtain an expression for the displacement component u_2 . Show that it is impossible to obtain the prescribed strain ϵ_{12} from the obtained expressions for u_1 and u_2 .
2. Consider the following 2D stress distribution on a body:

$$\sigma_{11} = -e^{x_1} \sin x_2, \quad \sigma_{22} = e^{x_1} \sin x_2, \quad \sigma_{12} = -e^{x_1} \cos x_2$$

- (a) Show that this stress distribution satisfies the equation of equilibrium when no body forces are present.
- (b) Assuming isotropic linear elastic material, find the corresponding 2D strains ϵ_{11} , ϵ_{12} , and ϵ_{22} .
- (c) Show that these strains *do satisfy* the compatibility equation

$$\frac{\partial^2 \epsilon_{11}}{\partial x_2^2} + \frac{\partial^2 \epsilon_{22}}{\partial x_1^2} = 2 \frac{\partial^2 \epsilon_{12}}{\partial x_1 \partial x_2}$$

- (d) Integrate the ϵ_{11} strains to obtain an expression for the displacement component u_1 and the ϵ_{22} strains to obtain an expression for the displacement component u_2 . Show that it is possible to obtain the prescribed strain ϵ_{12} from the obtained expressions for u_1 and u_2 .
3. Comment on the differences between cases 1 and 2. Does a stress field that satisfies conservation of linear and angular momentum necessarily make physical sense? Please elaborate.

Problem 2

A long, thick-walled pipe with inner radius a and outer radius b is embedded in the ground. The pipe contains fluid which causes a uniform internal pressure p_i as shown in Fig. 1. Assuming that the ground is a rigid rock, determine:

1. The stresses and displacements in the pipe in terms of a , b , E , and p_i for $\nu = 1/3$, where E and ν are the Young's modulus and Poisson's ratio respectively. **Note:** You may solve this with or without the use of an Airy Stress Function, either is ok. A good exercise is to do both!
2. The stresses at the outer boundary of the pipe (at $r = b$).

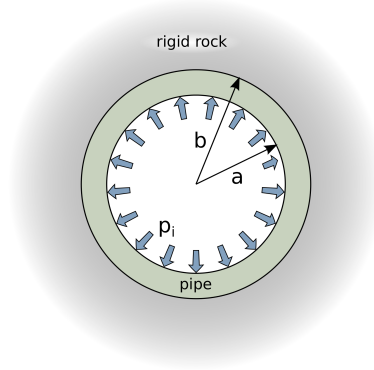


Figure 1: Schematic of a pressurized circular cylinder constrained on its exterior surface by a rigid body.

Problem 3

An infinite plate with a circular hole of radius a is subjected to a uniaxial stress state characterized by the stress σ_∞ at the infinity as shown in Fig. 2.

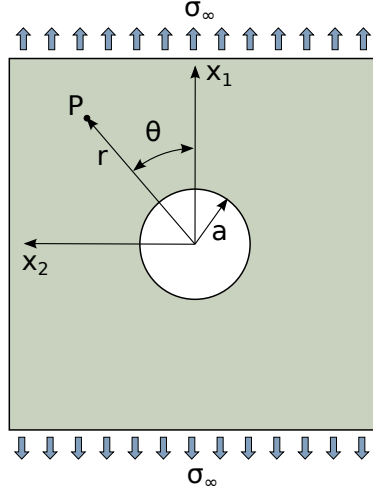


Figure 2: Schematic of an Infinite Plate with a hole subjected to far field uniaxial tension.

1. Express the boundary conditions for the stress tensor, in *polar components*, at $r = a$.
2. Express the boundary conditions for the stress tensor, in *Cartesian components*, as $r \rightarrow \infty$.
3. By transforming the stress components into polar coordinates, show that the latter become

$$\sigma_{rr} = \frac{1}{2}\sigma_\infty(1 + \cos 2\theta), \quad \sigma_{\theta\theta} = \frac{1}{2}\sigma_\infty(1 - \cos 2\theta), \quad \sigma_{r\theta} = -\frac{1}{2}\sigma_\infty \sin 2\theta \quad \text{as } r \rightarrow \infty$$

4. Using the definition of the Airy stress potential in polar coordinates for the problem with no body forces, integrate the expression for $\sigma_{\theta\theta}$ as $r \rightarrow \infty$ accordingly to obtain the following expression for the potential ϕ as $r \rightarrow \infty$.

$$\phi = \frac{1}{4}\sigma_\infty(1 - \cos 2\theta)r^2 + C_1(\theta)r + C_2(\theta) \quad \text{as } r \rightarrow \infty$$

5. Using this expression and the definition of σ_{rr} in terms of the potential, show that

$$\sigma_{rr} = \frac{1}{2}\sigma_\infty(1 + \cos 2\theta) + F(C_1(\theta), C_2(\theta)) \quad \text{as } r \rightarrow \infty$$

6. From the previous expression, it can be seen that the boundary conditions at $r \rightarrow \infty$ can be satisfied if we choose $C_1(\theta) = C_2(\theta) = 0$. Thus, the boundary conditions for both $\sigma_{\theta\theta}$ and σ_{rr} at $r \rightarrow \infty$ are satisfied by $\phi = \frac{1}{4}\sigma_\infty(1 - \cos 2\theta)r^2$ as $r \rightarrow \infty$. Show that this particular choice of ϕ also satisfies the boundary conditions for $\sigma_{r\theta}$ at $r \rightarrow \infty$.

7. This motivates us to look for solutions in the form of the following subset of Mitchell's solution:

$$\phi = \phi_0(r) + \phi_2(r) \cos 2\theta$$

with

$$\phi_0(r) = a_0 \ln r + b_0 r^2 + c_0 r^2 \ln r + d_0 \quad \text{and} \quad \phi_2(r) = a_2 r^2 + b_2 r^4 + c_2 r^{-2} + d_2$$

Determine the values for the constants so that all boundary conditions are satisfied.

8. Using the previously obtained functional, show that the stresses are

$$\begin{aligned}\sigma_{rr}(r, \theta) &= \frac{\sigma_{\infty}}{2} \left(1 - \frac{a^2}{r^2} \right) + \frac{\sigma_{\infty}}{2} \left(1 + 3\frac{a^4}{r^4} - 4\frac{a^2}{r^2} \right) \cos 2\theta \\ \sigma_{\theta\theta}(r, \theta) &= \frac{\sigma_{\infty}}{2} \left(1 + \frac{a^2}{r^2} \right) - \frac{\sigma_{\infty}}{2} \left(1 + 3\frac{a^4}{r^4} \right) \cos 2\theta \\ \sigma_{r\theta}(r, \theta) &= -\frac{\sigma_{\infty}}{2} \left(1 - 3\frac{a^4}{r^4} + 2\frac{a^2}{r^2} \right) \sin 2\theta\end{aligned}$$

9. At what points is the stress concentration higher? How much is the stress concentration factor for this problem? Draw the schematics.

Problem 4

An infinite plate with a circular hole of radius a is subjected to a uniform shear stress τ at the infinity as shown in Fig. 3.

1. Determine the stress distribution in the plate as a function of the **given** polar coordinates r and θ (measure r from the center of the hole and θ from the horizontal line in the counter-clockwise direction).
2. Find the maximum tensional (positive) and compressional (negative) hoop stress at the edge of the hole and determine the stress concentration factor.

Hint: The far field shear stress can be expressed as the superposition of a normal biaxial stress field with the appropriate choice of coordinate axes. Using these, you may solve this problem using superposition and the solution to Problem 3!

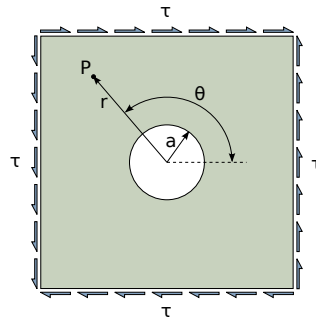


Figure 3: Schematic of an infinite plate with a hole subjected to uniform far field shear stress τ